

The model. A well-mixed population follows a unanimity q -voter rule. An external controller samples q_c agents once per round, chooses NOOP or ADVOCATE using a soft stochastic policy, and, if it advocates target A , replaces one of the q social-input slots in exactly b of the N microscopic update positions selected uniformly in advance.

1. State, timing, and variables

Simple picture. Agents update one at a time; the controller acts only once per population round. In the binary theory, A is the controller target and B the alternative. At the beginning of round k ,

$$N_k = n \in \{0, \dots, N\}, \quad x_k = n/N,$$

where n is the number of A supporters. One round contains exactly N focal-update opportunities.

Symbol	Meaning
N	population size and number of microscopic positions per round
q	number of social inputs seen by a focal agent
q_c	number of agents sensed by the controller at the round start
b	exact number of controlled positions in an advocacy round; $c = b/N$
Y_k	number of target supporters found in the controller sample
U_k	round action: NOOP or ADVOCATE
K_0, K_1	one-update kernels without / with controller advocacy
R_0, R_1	whole-round kernels under NOOP / ADVOCATE

The round protocol is

$$N_k \rightarrow Y_k \rightarrow U_k \rightarrow \underbrace{N \text{ microscopic updates}}_{\text{no new controller decision}} \rightarrow N_{k+1}.$$

If $U_k = \text{ADVOCATE}$, a set $S_k \subset \{1, \dots, N\}$ with $|S_k| = b$ is drawn uniformly *before* the round. Choosing update positions, not fixed agent identities, preserves exchangeability.

2. Population rule and microscopic kernels

Simple picture. The focal agent switches only when all q inputs unanimously support the opposite opinion. At a controlled position, the controller simply occupies one social-input slot and says A ; the focal response rule itself is unchanged.

Ordinary update:

$$K_0(n+1 | n) = \frac{N-n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}}, \quad K_0(n-1 | n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}}.$$

The diagonal term completes the row to one. Controlled update:

$$K_1(n+1 | n) = \frac{N-n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}}, \quad K_1(n-1 | n) = 0.$$

Again the diagonal completes the row. The controlled kernel is asymmetric by design; transfer entropy does not require microscopic reversibility.

3. Controller: finite sensing and a soft round decision

Simple picture. The unanimity rule belongs to the population. The sigmoid belongs only to the controller and decides whether the budget b is activated for the next round.

The controller samples q_c agents without replacement:

$$S(y | n) = \Pr(Y_k = y | N_k = n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}.$$

Given the measured target fraction y/q_c ,

$$\Pr(U_k = \text{ADVOCATE} | Y_k = y) = \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Thus the exact probability of advocacy at true state n is

$$a_n = \Pr(\text{ADVOCATE} | n) = \sum_y S(y | n) \sigma \left[\beta \left(\theta - \frac{y}{q_c} \right) \right].$$

For $q_c = 2$ and $\theta = 1/2$, this simplifies exactly to

$$a(x) = \frac{1}{2} + \frac{1}{2} \tanh \left(\frac{\beta}{4} \right) (1 - 2x).$$

For the illustrative choice $\beta = 4$, $a(x) = 0.880797 - 0.761594x$.

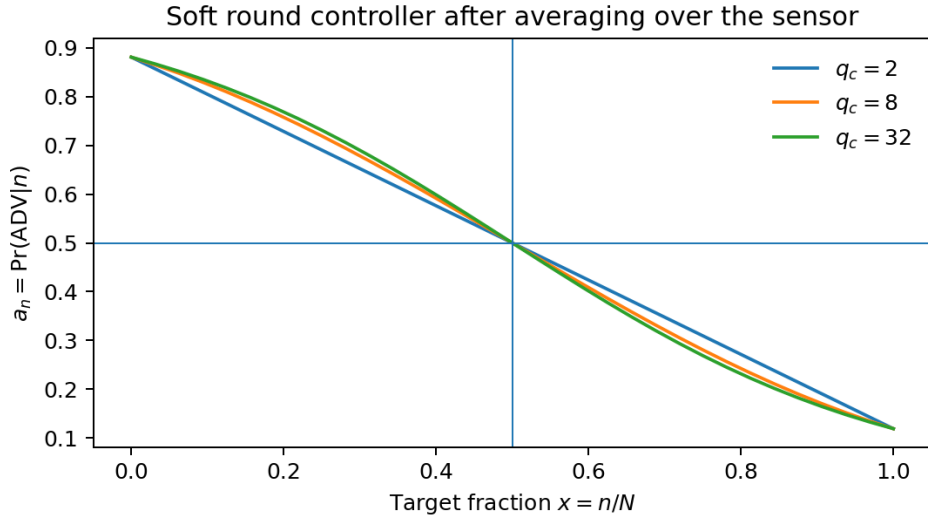


Figure 1. State-conditioned advocacy probability after averaging over the sensor for $N = 32$, $\theta = 1/2$, $\beta = 4$. q_c changes sensing fidelity; it does not change the unanimity population rule.

Parameter roles. q = social interaction order; q_c = sensing resource; θ = intervention trigger; β = controller decision stochasticity; $c = b/N$ = actuation resource.

4. Whole-round kernels

Simple picture. K_0, K_1 describe one focal update, whereas transfer entropy is measured from one round to the next. We therefore propagate the microscopic process through all N positions.

Under NOOP,

$$R_0 = K_0^N.$$

Under ADVOCATE, exactly b positions are controlled. Let $F_{r,j}$ be the transition matrix after r positions with j controlled positions already used. Starting from $F_{0,0} = I$,

$$F_{r+1,j+1} = \frac{b-j}{N-r} F_{r,j} K_1, \quad F_{r+1,j} = \frac{N-b-r+j}{N-r} F_{r,j} K_0,$$

so that

$$R_1 = F_{N,b}.$$

This is exact and averages over all uniformly preallocated control schedules without enumerating them.

5. Exact finite- N transfer entropy

Simple picture. Fix the current population state n . If knowing whether the controller chose ADVOCATE or NOOP changes the distribution of the next-round state, the action carries information into the population dynamics.

The action-marginalized next-state law is

$$M_n(m) = [1 - a_n]R_0(m | n) + a_n R_1(m | n).$$

The exact state-local transfer entropy is

$$T(n) = [1 - a_n] \sum_m R_0(m | n) \log_2 \frac{R_0(m | n)}{M_n(m)} + a_n \sum_m R_1(m | n) \log_2 \frac{R_1(m | n)}{M_n(m)}.$$

Equivalently,

$$T(n) = \text{JS}_{a_n}(R_0(\cdot | n), R_1(\cdot | n)),$$

a weighted Jensen-Shannon divergence between the exact no-control and advocacy round kernels. This is already an exact finite-population theoretical result: once $(N, q, q_c, b, \beta, \theta)$ are fixed, there is no empirical MI estimator in the calculation.

If $P_k(n)$ is the distribution of states at the start of round k ,

$$\mathcal{T}_k = I(U_k; N_{k+1} | N_k) = \sum_n P_k(n) T(n).$$

For a K -round experiment, the natural finite-horizon rate is

$$\overline{\mathcal{T}}^{(K)} = K^{-1} \sum_{k=0}^{K-1} \mathcal{T}_k.$$

The pure unanimity voter has absorbing consensus states, so long-time stationary TE can vanish after consensus. For comparison with finite LLM episodes, $T(n)$, $\mathcal{T}_k$, and $\overline{\mathcal{T}}^{(K)}$ are therefore more informative.

Two ingredients are required for positive TE. The soft policy provides action variation at fixed state through $0 < a_n < 1$. The population kernels provide dynamical separation through $R_0 \neq R_1$. If either ingredient disappears, $T(n) = 0$.

6. Closed large- N approximation

Simple picture. To understand how control should scale with q and b/N , approximate the count fraction $x = n/N$ as continuous and compare the mean drift with and without advocacy.

The ordinary and controlled drifts are

$$f_0(x) = (1 - x)x^q - x(1 - x)^q, \quad f_1(x) = (1 - x)x^{q-1}.$$

Their difference is

$$\Delta f_q(x) = x^{q-1}(1 - x)^2 + x(1 - x)^q.$$

With $c = b/N$, an advocacy round changes the mean drift by approximately $c\Delta f_q(x)$. The ordinary local noise coefficient is

$$\nu_0(x) = (1 - x)x^q + x(1 - x)^q - f_0(x)^2.$$

A second-order Jensen-Shannon expansion gives

$$T_{\text{MF}}(x) \approx \frac{a(x)[1 - a(x)]}{2 \ln 2} \frac{Nc^2[\Delta f_q(x)]^2}{\nu_0(x)}.$$

Hence the characteristic weak-control scaling is $T_{\text{MF}} \propto a(1 - a)b^2/N$, modulated by the q -dependent response-to-noise ratio.

7. Exact curves: what the classical theory predicts

Simple picture. The following are theory curves, not simulations. They use the exact $K_0, K_1 \rightarrow R_0, R_1 \rightarrow T(n)$ construction for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$, while varying q and the round budget b .

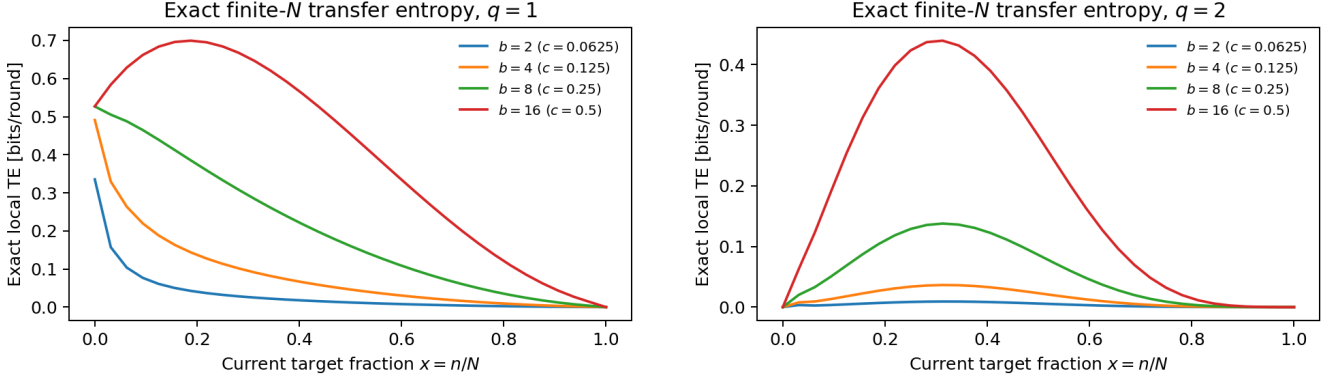


Figure 2. Exact state-local TE. For $q = 1$ (left), advocacy can seed target support even from $x = 0$, so the boundary channel can be strong. For $q = 2$ (right), $T(0) = 0$: one controller input still requires one ordinary A supporter to complete unanimity. The information channel therefore peaks in the interior.

8. The analytically stronger special case $q = 1$

For $q = 1$, ordinary copying has zero mean drift, $f_0(x) = 0$, whereas a controlled position gives $f_1(x) = 1 - x$. With exactly b controlled positions,

$$\mathbb{E}[x_{k+1} \mid x_k = x, \text{ADVOCATE}] = 1 - (1 - x) \left(1 - \frac{1}{N}\right)^b,$$

while $\mathbb{E}[x_{k+1} \mid x_k = x, \text{NOOP}] = x$. Therefore

$$\Delta\mu_{q=1}(x) = (1 - x) \left[1 - \left(1 - \frac{1}{N}\right)^b\right] \rightarrow (1 - x)(1 - e^{-c})$$

when $N \rightarrow \infty$ with $b = cN$.

This creates a qualitative boundary distinction:

$$q = 1 : T(0) > 0 \text{ for } b > 0, \quad q \geq 2 : T(0) = 0.$$

9. Control-budget dependence and compact comparison sheet

Simple picture. The final theoretical question is how much information can be injected as the round budget $c = b/N$ increases. The exact theory predicts a small-budget quadratic regime for $q \geq 2$, followed by saturation as the two round kernels become increasingly distinguishable.

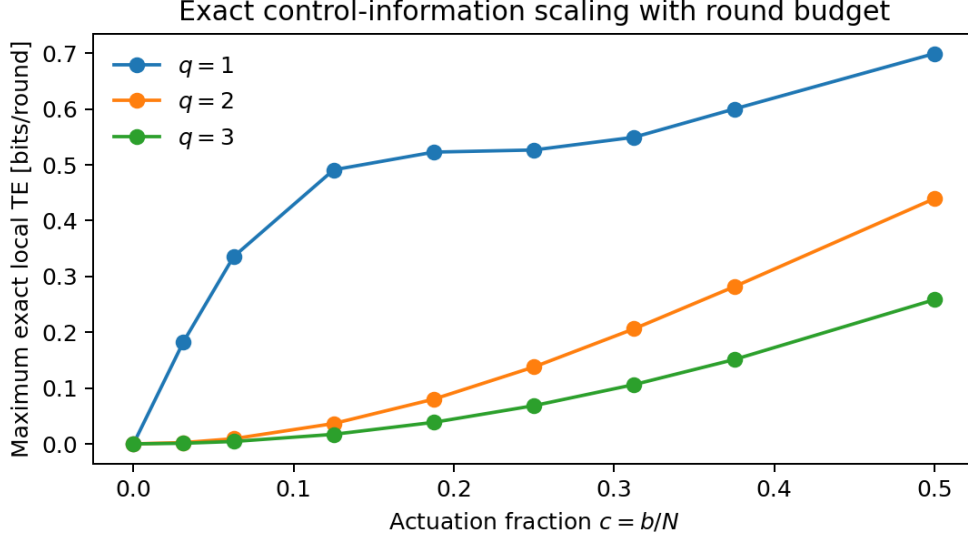


Figure 3. Maximum exact state-local TE over population states versus $c = b/N$ for $N = 32$, $q_c = 2$, $\theta = 1/2$, $\beta = 4$. For $q \geq 2$ the low-budget behavior is consistent with the c^2 scaling of the mean-field approximation. $q = 1$ departs earlier because advocacy opens otherwise inaccessible boundary transitions.

Object	Theoretical expression
Sensor	$S(y n) = \binom{n}{y} \binom{N-n}{q_c-y} / \binom{N}{q_c}$
Soft controller	$a_n = \sum_y S(y n) \sigma[\beta(\theta - y/q_c)]$
Population response	unanimity q -voter, encoded by K_0, K_1
Round dynamics	$R_0 = K_0^N, R_1 = F_{N,b}$
Exact local TE	$T(n) = JS_{a_n}[R_0(\cdot n), R_1(\cdot n)]$
Episode/round TE	$\mathcal{T}_k = \sum_n P_k(n) T(n)$
Mean-field separation	$\Delta f_q = x^{q-1}(1-x)^2 + x(1-x)^q$
Weak-control TE	$T_{MF} \approx \frac{a(1-a)}{2 \ln 2} \frac{N c^2 \Delta f_q^2}{\nu_0}$

What is fixed in this benchmark. Population response = unanimity q -voter; sensing = hypergeometric sample of size q_c ; controller decision = soft sigmoid; timing = once per round; actuation = exactly b preallocated positions, each replacing one of the q social inputs with advocacy for A . The LLM simulations can now be compared against these classical predictions at matched $(N, q, q_c, b, \beta, \theta)$.

10. Current as a finite-horizon control observable

Simple picture. Transfer entropy asks whether the controller action changes the distribution of the next state. A current asks a complementary question: *in which direction, and by how much, did the population actually move?* For the controller target A , define the one-round current

$$J_k = N_{k+1} - N_k.$$

Positive current means net motion toward A ; negative current means net motion away from A . Over a K -round episode,

$$J^{(K)} = N_K - N_0 = \sum_{k=0}^{K-1} (N_{k+1} - N_k).$$

The telescoping identity is useful: the net current is available directly from the initial and final target counts, even though the underlying trajectory may contain many back-and-forth switches.

At fixed current state n and action $u \in \{0, 1\}$, with $u = 0$ for NOOP and $u = 1$ for ADVOCATE, the exact one-round mean and variance follow directly from the whole-round kernels:

$$\mu_{J,u}(n) = \sum_m (m - n) R_u(m | n),$$

$$V_{J,u}(n) = \sum_m (m - n)^2 R_u(m | n) - \mu_{J,u}(n)^2.$$

Thus no new stochastic model is introduced: current statistics are another readout of the same exact R_0, R_1 theory used for transfer entropy.

Separation of roles. $T(n)$ measures action-conditioned distributional separation and is unsigned. $\mu_{J,1}(n) - \mu_{J,0}(n)$ measures the signed response toward the controller target. The two quantities answer different but complementary control questions.

11. Closed-form current response for $q = 1$

Simple picture. The linear voter case is analytically special. Ordinary copying has zero mean drift, while each controlled update creates a one-sided opportunity to recruit a non-target supporter.

For $q = 1$,

$$\mathbb{E}[N_{k+1} | N_k = n, \text{NOOP}] = n, \quad \mu_{J,0}^{(q=1)}(n) = 0.$$

With exactly b controlled positions in the round,

$$\mathbb{E}[N_{k+1} | N_k = n, \text{ADVOCATE}] = N - (N - n) \left(1 - \frac{1}{N}\right)^b,$$

so the exact finite- N one-round current is

$$\mu_{J,1}^{(q=1)}(n) = (N - n) \left[1 - \left(1 - \frac{1}{N}\right)^b\right].$$

Equivalently, for $x = n/N$,

$$\Delta\mu_x^{(q=1)}(x) = (1 - x) \left[1 - \left(1 - \frac{1}{N}\right)^b\right] \xrightarrow[N \rightarrow \infty]{b=cN} (1 - x)(1 - e^{-c}).$$

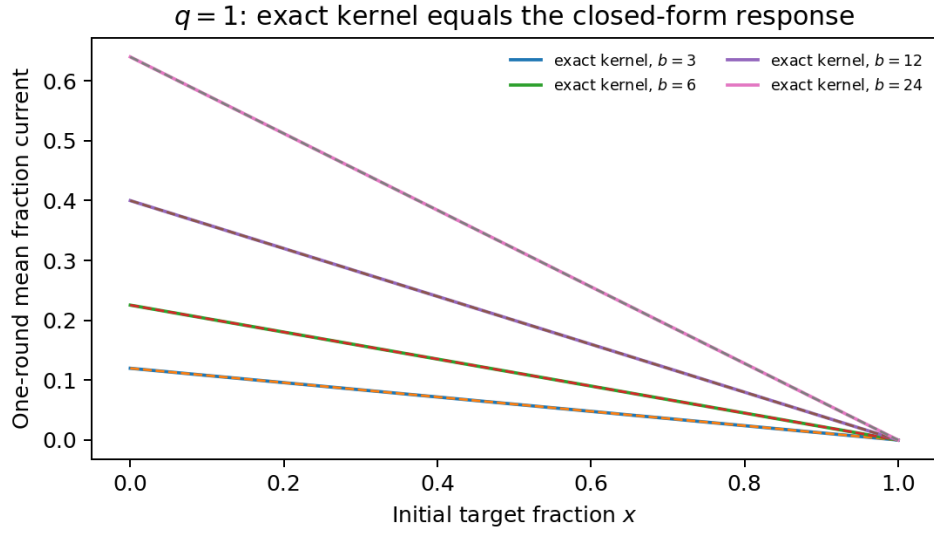


Figure 4. Exact one-round target-fraction current for $N = 24$, $q = 1$, and several budgets. Solid curves are obtained from the exact round kernel R_1 ; dashed curves are the scalar closed form above. Their coincidence is an internal theory check, not a fit.

12. Exact finite-horizon mean and variance

Simple picture. The simulations consist of complete episodes with a fixed number of rounds. The appropriate theoretical comparator is therefore also finite-horizon, rather than a stationary long-time current.

Let $\mathcal{R}$ denote the whole-round kernel used by the experiment: R_0 for always-NOOP, R_1 for always-ADVOCATE, or the row-wise feedback mixture

$$R(m | n) = [1 - a_n]R_0(m | n) + a_n R_1(m | n)$$

for a stochastic controller. If $P_0(n_0)$ is the initial target-count distribution, then after K rounds

$$P(N_K = m | N_0 = n_0) = [\mathcal{R}^K]_{n_0 m}.$$

Since $J^{(K)} = N_K - N_0$, the exact finite-horizon moments are

$$\mathbb{E}[J^{(K)}] = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0),$$

$$\mathbb{E}[(J^{(K)})^2] = \sum_{n_0, m} P_0(n_0) [\mathcal{R}^K]_{n_0 m} (m - n_0)^2,$$

$$\text{Var}(J^{(K)}) = \mathbb{E}[(J^{(K)})^2] - \mathbb{E}[J^{(K)}]^2.$$

These are exact finite- N , finite- K expressions. They require matrix propagation, but no Monte Carlo approximation.

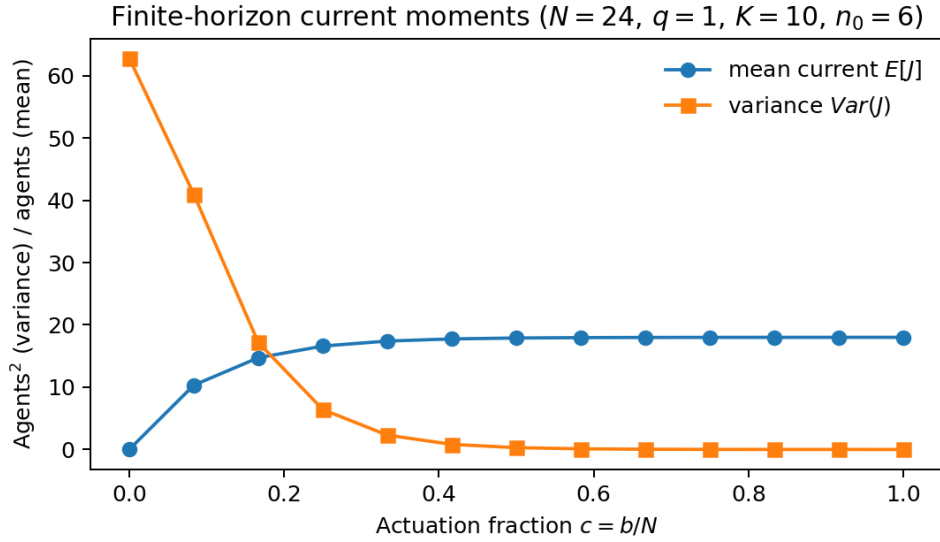


Figure 5. Exact finite-horizon mean current and current variance as the advocacy budget increases, for an illustrative $N = 24, q = 1, K = 10$ process beginning at $n_0 = 6$. The mean quantifies net steering; the variance quantifies trajectory-to-trajectory fluctuation expected even under the same macroscopic initial state.

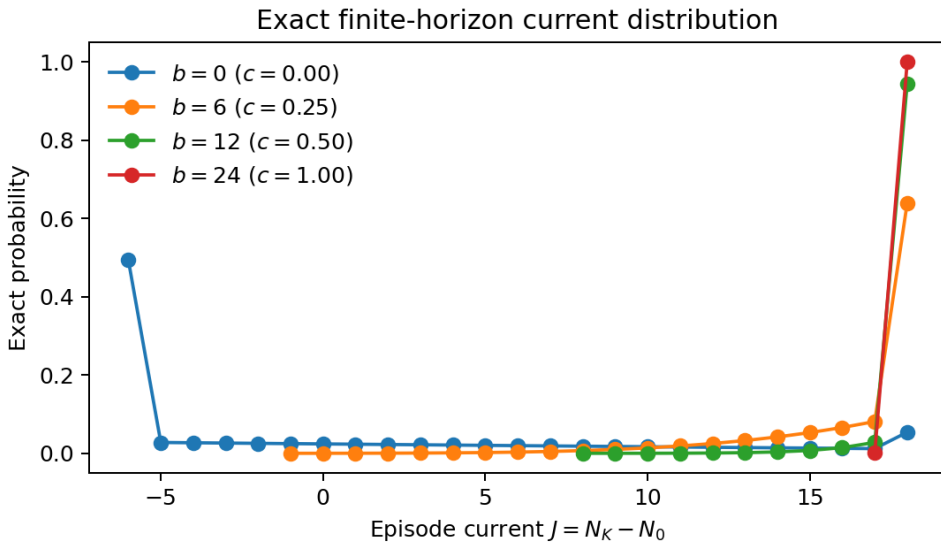


Figure 6. Exact distribution of the episode current $J = N_K - N_0$ for selected budgets in the same finite-horizon example. Increasing actuation shifts probability mass toward positive target current and changes the width of the response distribution.

13. Fano-like dispersion, inverse-Fano precision, and SNR^2

Simple picture. Mean current alone does not tell us whether steering is reproducible. Two control settings can produce the same average displacement but very different fluctuations across repeated episodes. The current variance therefore provides the natural second ingredient.

From the finite-horizon mean $\mu_J = \mathbb{E}[J]$ and variance $V_J = \text{Var}(J)$, define

$$F_J^{\text{disp}} = \frac{V_J}{|\mu_J|}$$

as a Fano-like dispersion, and

$$P_J^{\text{Iri}} = \frac{|\mu_J|}{V_J} = \frac{1}{F_J^{\text{disp}}}$$

as the inverse-Fano current-precision convention used in the stochastic-thermodynamics discussion of current precision. We keep both names explicit because the word ‘‘Fano factor’’ is used in opposite directions across literatures.

A particularly transparent dimensionless precision statistic is

$$\text{SNR}_J^2 = \frac{\mu_J^2}{V_J}.$$

It is the squared signal-to-noise ratio of the finite-horizon current: large values indicate that the directed displacement is large compared with stochastic fluctuations.

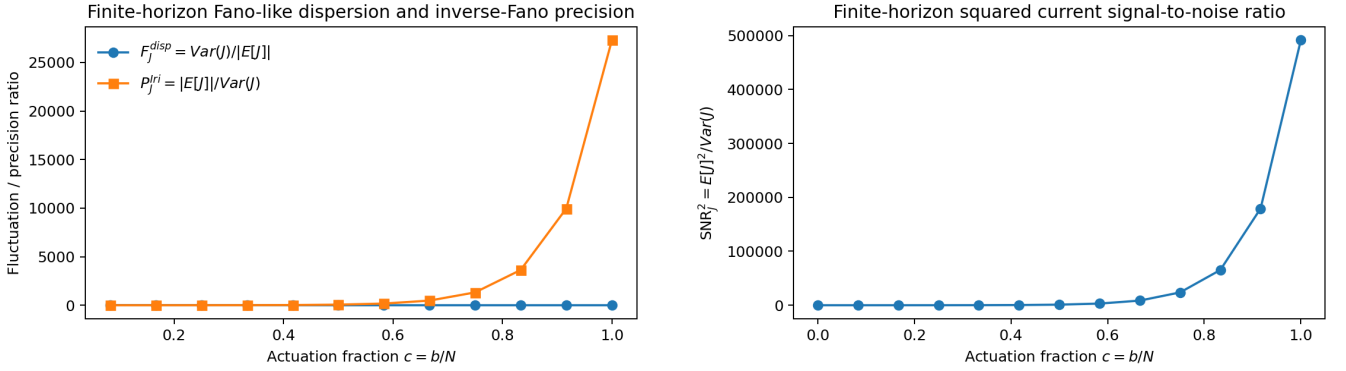


Figure 7. Exact finite-horizon fluctuation summaries for the illustrative $N = 24$, $q = 1$, $K = 10$, $n_0 = 6$ process. Left: dispersion and its inverse precision. Right: squared current signal-to-noise ratio. These quantities are derived from the same exact current distribution shown in Fig. 6.

How these quantities should be compared with LLM simulations. For each repeated LLM episode compute $J^{(e)} = n_Z^{(e)}(K) - n_Z^{(e)}(0)$. Across repetitions of the same task/cell, estimate the empirical mean, variance, F_J^{disp} , P_J^{Iri} , and SNR_J^2 . The theory values above are then evaluated at the matched (N, q, b, K) and initial target-count distribution. The theory is not re-fitted to the LLM data.

14. Unified control summary

The controlled q -voter now provides two complementary exact finite-population reference channels:

$$\text{information channel: } T(n) = \text{JS}_{a_n}(R_0, R_1),$$

$$\text{response channel: } J^{(K)} = N_K - N_0.$$

The first quantifies how distinguishable the controlled and uncontrolled next-state laws are; the second quantifies signed population transport toward the target. Their finite-horizon statistics provide a direct reference for empirical multi-agent runs:

$$\mathbb{E}[J], \quad \text{Var}(J), \quad F_J^{\text{disp}}, \quad P_J^{\text{Iri}}, \quad \text{SNR}_J^2.$$

For $q = 1$, the one-round mean current has the scalar closed form in Sec. 11. For general q and finite horizon, the same quantities remain exact through R_0, R_1 and matrix propagation.

Theory anchors. Irisarri et al., *Stochastic Thermodynamics of Social Imitation beyond Energetics* (2026), motivates the nonlinear imitation mechanism. Horowitz and Sandberg, *Second-law-like inequalities with information and their interpretations* (2014), motivates directional information in repeated feedback. The controlled kernel here is not constrained to be microscopically reversible.